Selecting Dependent and Independent Variables

HDI Prediction System | Epic 4: Data Preprocessing and Labeling

Objective

Define which columns serve as the model's input features (independent variables) and which column is the prediction target (dependent variable), matching exactly what the Flask app's input form collects.

Independent Variables (Features, X)

Feature	Dataset Column (2021)	Form Field in app.py
Life Expectancy at Birth	Life Expectancy at Birth (2021)	life
Expected Years of Schooling	Expected Years of Schooling (2021)	expected
Mean Years of Schooling	Mean Years of Schooling (2021)	mean
GNI per Capita	Gross National Income Per Capita (2021)	income

Dependent Variable (Target, y)

The target is **Human Development Index (2021)** — a continuous score between 0 and 1 representing the country's overall human development level, which the model learns to predict from the four independent variables above.

Code: Building X and y

```
feature_cols = [
    'Life Expectancy at Birth (2021)',
    'Expected Years of Schooling (2021)',
    'Mean Years of Schooling (2021)',
    'Gross National Income Per Capita (2021)'
]
target_col = 'Human Development Index (2021)'

# Drop rows missing the target or any of the 4 features
model_df = df.dropna(subset=feature_cols + [target_col])

X = model_df[feature_cols]
y = model_df[target_col]

print(X.shape, y.shape)
```

Why These Four Features?

- They map directly onto UNDP's three official HDI dimensions: health (life expectancy), education (expected + mean years of schooling), and standard of living (GNI per capita).

- They are the exact four fields the Flask form (index.html) collects from the user, so the model's training features must match the app's input schema exactly.
- All four columns have very low missing-value counts (0-4 rows out of 195), making them reliable to train on without heavy imputation.

Train/Test Split (for model evaluation)

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Next Step

With X and y defined, the next step in the Development Phase is training the regression model and serializing it to HDI.pkl for the Flask app to load.